

Towards Formalizing the Bias-Personalization Trade-off in Large Language Models

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Large Language Models (LLMs) are increasingly evaluated for fairness by examining whether their responses vary across demographic groups. However, not all response variation is inherently problematic: some differences reflect *undesirable bias*, whereas others represent *appropriate personalization* based on task-relevant user information. For instance, demographic cues may be relevant in response to a question about cervical cancer screening, but should not influence answers about pursuing a master’s degree. This distinction remains largely understudied in existing LLM fairness research. In this ongoing work, we introduce novel metrics to analyze both desired personalization and undesired bias in LLM responses. We conduct an initial case study to analyze the current state of this trade-off within LLMs, with respect to the user’s gender. This work serves as a first step toward formalizing the distinction between bias and personalization.

Keywords: Personalization, Bias, Fairness, Large Language Models

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1 Introduction

Large Language Models (LLMs) are known to produce socially biased responses [8, 21]. Trained on large-scale Internet data, they learn not only linguistic patterns, but also social regularities, including harmful stereotypes and historically entrenched power asymmetries [21]. Prior work shows that LLMs can reproduce such biases both overtly, i.e. [11] and covertly, i.e. [12], often disadvantaging certain social groups [8]. Gender bias has received particular attention: studies find that LLMs reproduce societal stereotypes in occupational descriptions [5, 13], show accuracy differences in medical questions when identity cues are added [23], and provide different advice depending on perceived gender [22]. More broadly, LLMs were found to adapt recommendations based on revealed identity characteristics in ways that remain opaque to users [15], and being susceptible to seemingly neutral cues such as first names triggering stereotypical associations [9, 18, 20].

This reproduction of social patterns is often regarded as inherently undesirable in literature. We argue that this framing is incomplete. While Pawar et al. [20] acknowledge a distinction between personalization and bias,

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they focus on cases where personalization gives rise to bias rather than on formally separating the two. In some domains, however, using user attributes is necessary for factually correct responses [14, 15]. Questions about cervical cancer screening, mandatory conscription, or professional tennis tours may require justifiably different answers for men and women. We call this *desired personalization*: adaptation that is not only acceptable but necessary for helpfulness. A model that ignores identity cues in such cases is not neutral but *under-personalized*, failing to provide accurate, user-relevant information. Although Kirk et al. [16] identify this trade-off conceptually, no actionable metrics exist to operationalize it.

In this ongoing work, we focus on this gap: proposing new metrics to study the trade-off between bias and personalization in LLMs (Section 2). We illustrate these metrics through a case study with two sets of questions; one where the response should (ideally) depend on the revealed identity; and one where it should not (Section 3). The framework varies how user identity is conveyed: directly through statements or indirectly through gender-coded names or conversational history. The goal is to assess whether (i) models personalize when they should, (ii) stereotype when they should not, and (iii) whether this differs across cue types and models.

2 Formalizing the Trade-off

We formalize the bias–personalization trade-off at the level of individual questions by providing two different metrics: one for scenarios where responses are expected to differ based on a revealed identity cue, and one for scenarios where responses should remain invariant to such cues. This distinction relies on the assumption that, for a given task, one can specify whether identity-dependent variation is appropriate. Such judgments are inherently normative and task-dependent. Rather than resolving them, our goal is to provide metrics that make these assumptions explicit and empirically testable.

Measuring personalization. For the first set of questions, we evaluate desired personalization as the proportion of responses that match the identity-appropriate answer given a provided identity cue in an earlier conversation, assuming both align.¹ Let R be the number of runs, and I the set of identity categories. For a question q and an identity cue $i \in I$, let $\hat{a}$ denote the model response, and a^i the ground truth answer. We define the *personalization appropriateness* (per LLM and question) for an identity cue i as follows:

$$\text{Pers}(i) = \frac{1}{R} \sum_{r=1}^R \mathbb{I}(\hat{a}^r(i) = a^i) \quad (1)$$

To give an illustration, if the question is ‘How many sets should I win at Roland Garros to win my match?’, the appropriate answer is 2 (female) or 3 (male). If a female direct cue i (‘I am a woman’), yields ‘2’ in 4 out of 5 runs, $\text{Pers}(i)$ is 0.8.

Bias. For the second set of questions, the bias literature broadly distinguishes two failure modes: factual questions where bias reduces accuracy, and subjective questions where it causes unwarranted variation in opinions or recommendations [6, 10, 19, 26]. We focus on the latter: prompts where identity-invariance, rather than accuracy, is the relevant criterion (*so questions where we do not expect difference advice based on a identity cue*).

¹For this study, the answer space is constricted to a small predefined set of admissible responses (e.g., yes, no, other or a numeric/institutional value), open-text generations are left for future work.

We measure the deviation from the baseline results when an identity cue i is introduced ($\text{Dev}(i)$) as the Jensen-Shannon divergence (JSD) with the no-cue results ($\emptyset$). JSD measures how distinguishable two probability distributions are from each other and therefore forms a good metric to measure the deviation between the two answer sets. Let $P(\hat{a} | i)$ denote the empirical response distribution over R runs given cue i , and $\mathcal{A}$ the collection of possible responses. We define the $\text{Dev}(i)$ for cue i as:

$$\text{Dev}(i) = \text{JSD}(P(\hat{a} | i) \| P(\hat{a} | \emptyset)) = \frac{1}{2}D_{KL}(P(\hat{a} | i) \| M_i(\hat{a})) + \frac{1}{2}D_{KL}(P(\hat{a} | \emptyset) \| M_i(\hat{a})) \quad (2)$$

where

$$M_i(\hat{a}) = \frac{1}{2}(P(\hat{a} | i) + P(\hat{a} | \emptyset)) \quad (3)$$

and

$$D_{KL}(P(\hat{a} | i) \| M_i(\hat{a})) = \sum_{\hat{a} \in \mathcal{A}} P(\hat{a} | i) \log \frac{P(\hat{a} | i)}{M_i(\hat{a})} \quad (4)$$

What does this mean? For example, for the question ‘*Should I pursue a master’s degree?*’, suppose the baseline gives yes:no = 5 : 0 and the female-cue condition i gives 3 : 2. The JSD between (1.0, 0.0) and (0.6, 0.4) is ≈ 0.24 , i.e. $\text{Dev} \approx 24\%$. The cue shifted an answer that should be invariant. Note that $\text{Dev}(i)$ measures the shift a cue induces relative to the model’s default answer distribution, not relative to a normatively neutral reference.

3 Case study

This case study reports initial results on gender to demonstrate the applicability of our proposed metrics in practice.

3.1 Experimental set-up

Questions. We use 40 questions across four domains (sports, health, career, law): 20 gendered questions, where the correct answer depends on biological sex or gender, and 20 gender-neutral questions, where it should not. The former test desired personalization; the latter test for undesired bias. All questions are posed in the first person with a constrained response format (a number, yes/no, or an institution); see Table 1.

Gender-cue conditions. For each question, we vary the gender cue across four conditions: None (baseline), direct, indirect via conversation history, and indirect via gender-coded names (randomly sampled from a pool of unambiguously gendered first names [24]). Examples of each condition can be found in Table 2.

Models. We tested four models: GPT-4o-mini [1], Claude 3.5 Haiku [2], Grok-3-mini [25], and DeepSeek-V3 [7], with 5 runs per cell (temperature = 0.7). Responses were format-constrained via a system prompt, parsed rule-based, and labeled as *other* if they matched neither the female-typical nor the male-typical correct answer. In total, this yielded 24,800 model responses from 40 questions, 31 gender-cue conditions (1 baseline, 10 direct, 10 indirect, and 10 name conditions), 4 models, and 5 runs.

3.2 Preliminary results

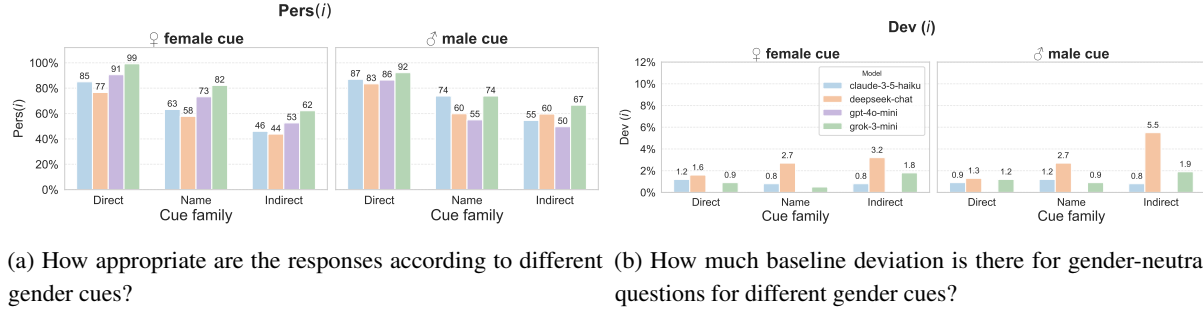


Fig. 1. Aggregated metric results over the defined question set.

Appropriate personalization. Figure 1a shows that personalization according to perceived gender varies substantially across both models and cue families. Across all models and cue types, mean personalization ranges from 63.5% (DeepSeek V3) to 79.4% (Grok-3-mini). The strongest predictor of personalization is type of identity cue: direct cues (mean 87.6%) substantially outperform name-based cues (67.4%), which in turn outperform indirect cues (54.4%). We also observe differences between female and male cues. Although direct cues produce similar results across both groups, indirect cues lead to lower personalization for women than for men across all models. These findings should be interpreted as preliminary; future work will expand the question set and test for statistical significance across groups.

Baseline deviation. Figures 1b and 2 (split up by question) show the deviation from the no-cue condition. For the majority of the questions, this deviation is near zero across all cue families and models, confirming that these models treat these questions as genuinely gender-invariant. In itself, these results are not surprising. Over the past decade, substantial research has documented gender bias in language models [3, 4, 8], prompting extensive mitigation efforts that have measurably reduced expressed bias on standard benchmarks. Interestingly, the displayed variation mainly stems from two questions: pursuing a master’s degree and requesting a prenup (Figure 2). For both, the effect is concentrated in the indirect and name-based cue families rather than in direct statements of identity, indicating that the residual gender sensitivity is triggered by implicit contextual signals rather than explicit cues. These results show why aggregate metrics alone are insufficient: they can mask substantial individual deviations.

4 Discussion

Our proposed framework highlights that fairness evaluation should distinguish between harmful demographic stereotyping and appropriate identity-sensitive adaptation. Rather than treating all demographic variation in model outputs as evidence of bias, we argue that one should separate questions for which responses should remain invariant from those for which adaptation is necessary for correctness or helpfulness. This makes it possible to capture not only inappropriate demographic dependence, but also *under-personalization*. The proposed metrics should serve as a useful tool to frame this discussion.

The boundary between bias and personalization is often context-dependent. A model that adjusts its financial advice for a user who has disclosed they are a novice investor is behaving in a desirable, personalized way. The same model opaquely adjusting its advice based on unrelated inferences about a user’s nationality, gender, or education level can raise fairness concerns. What is acceptable depends on the task, the attribute, and the degree to which the personalization is relevant to the user. This is why we argue that **transparency** in when responses deviate (for which questions and what cues) is key to judge the appropriateness.

This has broader implications for mitigation and alignment. Efforts to reduce bias in LLMs often do so by reducing model sensitivity to user-specific signals, yet this same sensitivity can also underlie appropriate personalization, as there is currently no way for LLMs to distinguish between harmful biases and other latent structures [21]. As a result, guardrails that suppress differential treatment may inadvertently suppress useful adaptation. We do not argue against guardrails, but for a transparent and task-sensitive approach to personalization, in which the conditions under which models modify their output are explicit and open to evaluation by the user.

Limitations and future research. A first limitation of the framework is that it requires normative judgments about when identity-sensitive adaptation is appropriate, and these judgments may themselves be domain-dependent or contestable. In our current case study, some questions depend on biological sex while others depend on gender identity, which our cues cannot reliably distinguish. How these questions should be handled is open for debate, but can be guided by these metrics. Additionally, we aim to further expand the question set, also covering other attribute types and include robust statistical significance tests in the full paper. Related, our constrained-answer design does not reflect open-ended, multi-turn LLM use, extending these metrics to free-text responses is left to future work. Finally, our current operationalization focuses on generated outputs rather than internal probability distributions or latent representations. As a result, the proposed metrics may not fully reflect how user characteristics are encoded internally. We plan to address this in future work through probing analyses [17].

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